Module 3: Linearization and Stability Analysis of Multicopter Dynamics

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1 Review: Nonlinear Systems and Linearization

A **physical system** is a real system, such as a car, an aircraft, or the climate. A **mathematical system** is a mathematical description of a physical system. The process of describing a physical system using mathematics is called **modeling**, and the resulting description is called the **system model**.

Example 1.1. Consider a mass-damper-spring (MCK) system. Using Newton's Second Law, the system can be described by

$$m\ddot{q} + c\dot{q} + kq = f, \tag{1}$$

where q is the displacement, m is the mass, c is the damper coefficient, k is the spring constant, and f is the applied force. The differential equation (1) is a model of the physical MCK system.

Physical systems are often modeled by differential equations, and these models are *typically nonlinear*. However, analysis and control design for general nonlinear models can be difficult, while linear models admit powerful tools (stability tests, controller synthesis, observers). Therefore, a central objective in control is to obtain a **linear or linearized approximation** of a nonlinear system that is valid in a neighborhood of an equilibrium or a trajectory.

1.1 Nonlinear State-Space Form

A nonlinear system can be written in state-space form as

$$\dot{x} = f(x, u, t), \tag{2}$$
$$y = h(x, u, t), \tag{3}$$

where $x \in \mathbb{R}^n$ is the state, $u \in \mathbb{R}^m$ is the input, $y \in \mathbb{R}^p$ is the output, and f and h are vector-valued functions. If f and h do not explicitly depend on t, the system is **time-invariant** and can be written as $\dot{x} = f(x, u)$.

1.2 Equilibrium and Linearization

For the time-invariant system $\dot{x} = f(x, u)$, a pair $(x_\mathrm{e}, u_\mathrm{e})$ is an **equilibrium state-input pair** if

$$f(x_\mathrm{e}, u_\mathrm{e}) = 0. \tag{4}$$

To study behavior near $(x_\mathrm{e}, u_\mathrm{e})$, define perturbations

$$x = x_\mathrm{e} + \xi, \qquad u = u_\mathrm{e} + v, \tag{5}$$

where ξ and v are the state and input perturbations. Using a first-order Taylor expansion about $(x_\mathrm{e}, u_\mathrm{e})$ gives

$$f(x_\mathrm{e} + \xi, u_\mathrm{e} + v) \approx f(x_\mathrm{e}, u_\mathrm{e}) + \frac{\partial f}{\partial x}\Big|_{(x_\mathrm{e}, u_\mathrm{e})} \xi + \frac{\partial f}{\partial u}\Big|_{(x_\mathrm{e}, u_\mathrm{e})} v. \tag{6}$$

Since $f(x_\mathrm{e}, u_\mathrm{e}) = 0$, the resulting **linearized perturbation dynamics** is

$$\dot{\xi} = A\xi + Bv, \tag{7}$$

where

$$A \triangleq \frac{\partial f}{\partial x}\Big|_{(x_{\mathrm{e}},u_{\mathrm{e}})}, \qquad B \triangleq \frac{\partial f}{\partial u}\Big|_{(x_{\mathrm{e}},u_{\mathrm{e}})}. \tag{8}$$

The approximation (7) is **local**: it is accurate only in a neighborhood of the equilibrium where higher-order terms are negligible. In this course, we will linearize the multicopter equations of motion about hover to obtain an LTI model that is suitable for stability analysis and feedback control design.

2 Nonlinear Multicopter Model (Review)

We consider a multicopter with inertial frame $\mathrm{F_A}$ (NED) and body-fixed frame $\mathrm{F_D}$. The equations of motion in state space form are

$$\dot{x} = f(x,u), \tag{9}$$

where

$$x \triangleq \begin{bmatrix} p \\ \dot{p} \\ q \\ \omega \end{bmatrix} \in \mathbb{R}^{12}, u \triangleq \begin{bmatrix} T \\ \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix} \in \mathbb{R}^4 \tag{10}$$

and

$$f(x,u) = \begin{bmatrix} \dot{p} \\ ge_3 - \mathcal{O}_{A/D}(q)e_3T \\ S(q)^{-1}\omega \\ J^{-1}(-\omega^\times J\omega + \tau) \end{bmatrix} \in \mathbb{R}^{12}, \tag{11}$$

2.1 Hover Equilibrium

The **hover equilibrium** corresponds to constant position, zero translational velocity, level attitude, and zero body angular velocity. Specifically, choose

$$\dot{p}_{\mathrm{e}} = 0, \tag{12}$$

$$\omega_{\mathrm{e}} = 0, \tag{13}$$

$$q_{\mathrm{e}} = 0 \text{ such that } \mathcal{O}_{A/D}(q_{\mathrm{e}}) = I_3, \tag{14}$$

and select the equilibrium input

$$T_{\mathrm{e}} = g, \tag{15}$$

$$\tau_{\mathrm{e}} = 0, \tag{16}$$

so that the translational and rotational accelerations are zero. Thus, for any constant hover position $p_{\mathrm{e}} \in \mathbb{R}^3$, the hover equilibrium state-input pair can be written as

$$x_{\mathrm{e}} = \begin{bmatrix} p_{\mathrm{e}} \\ 0 \\ 0 \\ 0 \end{bmatrix}, \qquad u_{\mathrm{e}} = \begin{bmatrix} g \\ 0 \\ 0 \\ 0 \end{bmatrix}, \tag{17}$$

which satisfies $f(x_{\mathrm{e}}, u_{\mathrm{e}}) = 0$.

2.2 Linearization About Hover: A and B Matrices

Let (x_e, u_e) denote the hover equilibrium from the previous subsection, with

$$q_e = 0, \quad \mathcal{O}_{A/D}(q_e) = I_3, \quad \dot{p}_e = 0, \quad \omega_e = 0, \quad T_e = g, \quad \tau_e = 0.$$

Define perturbations

$$x = x_e + \delta x, \qquad u = u_e + \delta u,$$

and write the linearized dynamics as

$$\delta\dot{x} = A\,\delta x + B\,\delta u, \qquad A \triangleq \left.\frac{\partial f}{\partial x}\right|_{(x_e,u_e)}, \qquad B \triangleq \left.\frac{\partial f}{\partial u}\right|_{(x_e,u_e)}. \tag{18}$$

Assume $q = \begin{bmatrix}\phi & \theta & \psi\end{bmatrix}^{\mathrm{T}}$ are the 3-2-1 Euler angles, so that for small angles

$$\mathcal{O}_{A/D}(q)e_3 \approx \begin{bmatrix}\theta \\ -\phi \\ 1\end{bmatrix}, \qquad \delta\big(\mathcal{O}_{A/D}(q)e_3\big) = \begin{bmatrix}\delta\theta \\ -\delta\phi \\ 0\end{bmatrix}. \tag{19}$$

Linearizing $\ddot{p} = ge_3 - \mathcal{O}_{A/D}(q)e_3\,T$ about hover yields

$$\delta\ddot{p} = -g\,\delta\big(\mathcal{O}_{A/D}(q)e_3\big) - e_3\,\delta T = \begin{bmatrix}-g\,\delta\theta \\ g\,\delta\phi \\ -\delta T\end{bmatrix}. \tag{20}$$

Moreover, since $q_e = 0$ and $\omega_e = 0$, we have $S(q_e)^{-1} = I_3$ and the kinematics linearize to

$$\delta\dot{q} = \delta\omega, \tag{21}$$

and the rotational dynamics linearize to

$$\delta\dot{\omega} = J^{-1}\delta\tau, \qquad \delta\tau \triangleq \begin{bmatrix}\delta\tau_1 & \delta\tau_2 & \delta\tau_3\end{bmatrix}^{\mathrm{T}}. \tag{22}$$

Let the state ordering be

$$\delta x = \begin{bmatrix}\delta p \\ \delta\dot{p} \\ \delta q \\ \delta\omega\end{bmatrix} \in \mathbb{R}^{12}, \qquad \delta u = \begin{bmatrix}\delta T \\ \delta\tau_1 \\ \delta\tau_2 \\ \delta\tau_3\end{bmatrix} \in \mathbb{R}^4.$$

Then the linearized matrices have the block form

$$A = \begin{bmatrix}0_{3\times3} & I_3 & 0_{3\times3} & 0_{3\times3} \\ 0_{3\times3} & 0_{3\times3} & A_{vq} & 0_{3\times3} \\ 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & I_3 \\ 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & 0_{3\times3}\end{bmatrix}, \qquad B = \begin{bmatrix}0_{3\times1} & 0_{3\times3} \\ B_{vT} & 0_{3\times3} \\ 0_{3\times1} & 0_{3\times3} \\ 0_{3\times1} & J^{-1}\end{bmatrix}, \tag{23}$$

where (with $q = \begin{bmatrix}\phi & \theta & \psi\end{bmatrix}^{\mathrm{T}}$)

$$A_{vq} = \frac{\partial(\delta\ddot{p})}{\partial(\delta q)} = \begin{bmatrix}0 & -g & 0 \\ g & 0 & 0 \\ 0 & 0 & 0\end{bmatrix}, \qquad B_{vT} = \frac{\partial(\delta\ddot{p})}{\partial(\delta T)} = \begin{bmatrix}0 \\ 0 \\ -1\end{bmatrix}. \tag{24}$$

If $J = \mathrm{diag}(J_1, J_2, J_3)$, then

$$J^{-1} = \mathrm{diag}\left(\frac{1}{J_1}, \frac{1}{J_2}, \frac{1}{J_3}\right). \tag{25}$$

Equivalently, the linearized perturbation dynamics can be read directly from (23):

$$\delta\dot{p} = \delta\dot{p}, \tag{26}$$

$$\delta\ddot{p} = \begin{bmatrix} -g\,\delta\theta \\ g\,\delta\phi \\ -\delta T \end{bmatrix}, \tag{27}$$

$$\delta\dot{q} = \delta\omega, \tag{28}$$

$$\delta\dot{\omega} = J^{-1}\delta\tau. \tag{29}$$

2.3 Linearization About Hover: Translational and Rotational Subsystems

Let $(x_\mathrm{e}, u_\mathrm{e})$ denote the hover equilibrium with

$$q_\mathrm{e} = 0, \quad \mathcal{O}_{A/D}(q_\mathrm{e}) = I_3, \quad \dot{p}_\mathrm{e} = 0, \quad \omega_\mathrm{e} = 0, \quad T_\mathrm{e} = g, \quad \tau_\mathrm{e} = 0.$$

Define perturbations

$$p = p_\mathrm{e} + \delta p, \quad \dot{p} = 0 + \delta\dot{p}, \quad q = 0 + \delta q, \quad \omega = 0 + \delta\omega, \quad u = u_\mathrm{e} + \delta u,$$

with $\delta u = \begin{bmatrix} \delta T & \delta\tau_1 & \delta\tau_2 & \delta\tau_3 \end{bmatrix}^\mathrm{T}$ and

$$q \triangleq \begin{bmatrix} \psi & \theta & \phi \end{bmatrix}^\mathrm{T}.$$

2.3.1 Translational Dynamics

The translational dynamics are

$$\dot{p} = \dot{p}, \qquad \ddot{p} = ge_3 - \mathcal{O}_{A/D}(q)e_3\,T.$$

Using 3-2-1 Euler angles with ordering $q = \begin{bmatrix} \psi & \theta & \phi \end{bmatrix}^\mathrm{T}$, and linearizing about $q_\mathrm{e} = 0$, we have the small-angle approximation

$$\mathcal{O}_{A/D}(q)e_3 \approx \begin{bmatrix} \theta \\ -\phi \\ 1 \end{bmatrix}, \qquad \delta\big(\mathcal{O}_{A/D}(q)e_3\big) = \begin{bmatrix} \delta\theta \\ -\delta\phi \\ 0 \end{bmatrix}.$$

Therefore, with $T_\mathrm{e} = g$, the linearized translational dynamics become

$$\delta\dot{p} = \delta\dot{p}, \tag{30}$$

$$\delta\ddot{p} = -g\,\delta\big(\mathcal{O}_{A/D}(q)e_3\big) - e_3\,\delta T = \begin{bmatrix} -g\,\delta\theta \\ g\,\delta\phi \\ -\delta T \end{bmatrix}. \tag{31}$$

Define the translational state and input as

$$x_{\mathrm{tr}} \triangleq \begin{bmatrix} \delta p \\ \delta \dot{p} \end{bmatrix} \in \mathbb{R}^6, \qquad u_{\mathrm{tr}} \triangleq \begin{bmatrix} \delta T \\ \delta q \end{bmatrix} = \begin{bmatrix} \delta T \\ \delta \psi \\ \delta \theta \\ \delta \phi \end{bmatrix} \in \mathbb{R}^4.$$

Then (30)–(31) can be written as

$$\dot{x}_{\mathrm{tr}} = A_{\mathrm{tr}} x_{\mathrm{tr}} + B_{\mathrm{tr}} u_{\mathrm{tr}}, \tag{32}$$

where

$$A_{\mathrm{tr}} = \begin{bmatrix} 0_{3\times 3} & I_3 \\ 0_{3\times 3} & 0_{3\times 3} \end{bmatrix}, \qquad B_{\mathrm{tr}} = \begin{bmatrix} 0_{3\times 1} & 0_{3\times 3} \\ B_{vT} & B_{vq} \end{bmatrix}, \tag{33}$$

with

$$B_{vT} = \frac{\partial(\delta \ddot{p})}{\partial(\delta T)} = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}, \tag{34}$$

$$B_{vq} = \frac{\partial(\delta \ddot{p})}{\partial(\delta q)} = \begin{bmatrix} 0 & -g & 0 \\ 0 & 0 & g \\ 0 & 0 & 0 \end{bmatrix}, \qquad (\delta q = \begin{bmatrix} \delta\psi & \delta\theta & \delta\phi \end{bmatrix}^{\mathrm{T}}). \tag{35}$$

2.3.2 Rotational Dynamics

The rotational dynamics are

$$\dot{q} = S(q)^{-1}\omega, \qquad \dot{\omega} = J^{-1}\left(-\omega^{\times} J\omega + \tau\right).$$

At hover, $q_{\mathrm{e}} = 0$ implies $S(q_{\mathrm{e}})^{-1} = I_3$, and $\omega_{\mathrm{e}} = 0$ implies $\omega^{\times} J\omega$ is second order and can be neglected in the linearization. Thus,

$$\delta \dot{q} = \delta\omega, \tag{36}$$

$$\delta\dot{\omega} = J^{-1}\delta\tau, \qquad \delta\tau \triangleq \begin{bmatrix} \delta\tau_1 \\ \delta\tau_2 \\ \delta\tau_3 \end{bmatrix}. \tag{37}$$

Define the rotational state and input as

$$x_{\mathrm{rot}} \triangleq \begin{bmatrix} \delta q \\ \delta\omega \end{bmatrix} \in \mathbb{R}^6, \qquad u_{\mathrm{rot}} \triangleq \delta\tau \in \mathbb{R}^3.$$

Then (36)–(37) can be written as

$$\dot{x}_{\mathrm{rot}} = A_{\mathrm{rot}} x_{\mathrm{rot}} + B_{\mathrm{rot}} u_{\mathrm{rot}}, \tag{38}$$

where

$$A_{\mathrm{rot}} = \begin{bmatrix} 0_{3\times 3} & I_3 \\ 0_{3\times 3} & 0_{3\times 3} \end{bmatrix}, \qquad B_{\mathrm{rot}} = \begin{bmatrix} 0_{3\times 3} \\ J^{-1} \end{bmatrix}, \tag{39}$$

and if $J = \mathrm{diag}(J_1, J_2, J_3)$, then

$$J^{-1} = \mathrm{diag}\left(\frac{1}{J_1}, \frac{1}{J_2}, \frac{1}{J_3}\right).$$

2.3.3 Remark: Cascade Structure at Hover

Equations (30)–(31) show that, to first order, the translational accelerations are driven by θ and ϕ (with no first-order dependence on ψ), while (36)–(37) show that the Euler angles are driven by body torques through the angular rates. This motivates cascaded control: an inner-loop attitude controller (using τ) and an outer-loop position controller (using T and attitude commands).

2.3.4 Structural Interpretation of the Linearized Model

The linearized hover dynamics reveal a clear double-integrator structure in both the translational and rotational subsystems.

From (30)–(31), the translational dynamics can be written componentwise as

$$\delta\ddot{p}_1 = -g\,\delta\theta, \tag{40}$$
$$\delta\ddot{p}_2 = g\,\delta\phi, \tag{41}$$
$$\delta\ddot{p}_3 = -\delta T. \tag{42}$$

Therefore,

- $(p_1, \dot{p}_1)$ is a double integrator driven by θ,
- $(p_2, \dot{p}_2)$ is a double integrator driven by ϕ,
- $(p_3, \dot{p}_3)$ is a double integrator driven by T.

In particular, yaw ψ does not appear in the translational dynamics to first order.

Similarly, from (36)–(37), the rotational dynamics are

$$\delta\dot{q} = \delta\omega, \tag{43}$$
$$\delta\dot{\omega} = J^{-1}\delta\tau, \tag{44}$$

which implies, componentwise,

$$\delta\ddot{\psi} = \frac{1}{J_3}\delta\tau_3, \tag{45}$$
$$\delta\ddot{\theta} = \frac{1}{J_2}\delta\tau_2, \tag{46}$$
$$\delta\ddot{\phi} = \frac{1}{J_1}\delta\tau_1. \tag{47}$$

Thus,

- each rotational channel is a double integrator driven by τ_i.

Key Insight. At hover, the multicopter linearizes into six coupled double integrators. The rotational dynamics are directly actuated by torques, while the translational dynamics are *indirectly actuated* through the attitude angles. This structural property motivates cascaded control: stabilize attitude first (inner loop), then regulate position (outer loop).

3 Stability of Linear Systems

Consider the linear time-invariant (LTI) system

$$\dot{x} = Ax, \tag{48}$$

where $x \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n\times n}$. The stability properties of (48) are completely determined by the matrix A.

3.1 Definitions

- The equilibrium $x = 0$ is **Lyapunov stable** if for every $\epsilon > 0$ there exists $\delta > 0$ such that

$$\|x(0)\| < \delta \quad \Rightarrow \quad \|x(t)\| < \epsilon \text{ for all } t \geq 0.$$

- The equilibrium is **asymptotically stable** if it is Lyapunov stable and

$$\lim_{t\to\infty} x(t) = 0 \quad \text{for all initial conditions.}$$

- The equilibrium is **exponentially stable** if there exist constants $c > 0$ and $\lambda > 0$ such that

$$\|x(t)\| \leq ce^{-\lambda t}\|x(0)\| \quad \text{for all } t \geq 0.$$

- The equilibrium is **unstable** if it is not Lyapunov stable.

3.2 Eigenvalue Test

For the LTI system (48), stability is determined by the eigenvalues of A.

- The system is **asymptotically (and exponentially) stable** if and only if

$$\mathrm{Re}(\lambda_i(A)) < 0 \quad \text{for all } i.$$

- The system is **unstable** if at least one eigenvalue satisfies

$$\mathrm{Re}(\lambda_i(A)) > 0.$$

- If all eigenvalues satisfy $\mathrm{Re}(\lambda_i(A)) \leq 0$ and any eigenvalue with zero real part is defective (nontrivial Jordan block), then the system is unstable.
- If all eigenvalues satisfy $\mathrm{Re}(\lambda_i(A)) \leq 0$ and all eigenvalues with zero real part are simple (no Jordan blocks), then the system is Lyapunov stable but not asymptotically stable.

3.3 Implication for Hover

The linearized hover model contains multiple integrators, that is, eigenvalues at the origin. Therefore, the open-loop hover equilibrium is not asymptotically stable. Small perturbations in position or attitude do not decay without feedback.

4 Open-Loop Stability of Translational and Rotational Subsystems

We now analyze the stability of the linearized hover dynamics by examining the eigenvalues of the translational and rotational subsystems separately.

4.1 Translational Subsystem

Recall that the translational state is

$$x_{\mathrm{tr}} \triangleq \begin{bmatrix} \delta p \\ \delta \dot{p} \end{bmatrix} \in \mathbb{R}^6,$$

and the linearized dynamics (ignoring inputs) are

$$\delta \dot{p} = \delta \dot{p}, \tag{49}$$

$$\delta \ddot{p} = \begin{bmatrix} -g\,\delta\theta \\ g\,\delta\phi \\ 0 \end{bmatrix}. \tag{50}$$

If we examine the *open-loop natural dynamics* (set $\delta T = 0$ and $\delta q = 0$), the translational subsystem reduces to

$$\delta \ddot{p} = 0. \tag{51}$$

In state-space form,

$$\dot{x}_{\mathrm{tr}} = A_{\mathrm{tr}} x_{\mathrm{tr}}, \qquad A_{\mathrm{tr}} = \begin{bmatrix} 0_{3\times3} & I_3 \\ 0_{3\times3} & 0_{3\times3} \end{bmatrix}. \tag{52}$$

The characteristic polynomial of A_{tr} is

$$\det(\lambda I - A_{\mathrm{tr}}) = \lambda^6.$$

Thus, the eigenvalues are

$$\lambda_i = 0, \qquad i = 1, \ldots, 6.$$

Conclusion (Translational Dynamics). The translational subsystem has six eigenvalues at the origin. Therefore:

- It is not asymptotically stable.
- It is not exponentially stable.
- It behaves as a collection of double integrators.

Small perturbations in position lead to constant velocity drift. Small perturbations in velocity lead to linear growth in position.

4.2 Rotational Subsystem

The rotational state is

$$x_{\text{rot}} \triangleq \begin{bmatrix} \delta q \\ \delta \omega \end{bmatrix} \in \mathbb{R}^6,$$

and the linearized dynamics (ignoring inputs) are

$$\delta \dot{q} = \delta \omega, \tag{53}$$

$$\delta \dot{\omega} = 0. \tag{54}$$

In state-space form,

$$\dot{x}_{\text{rot}} = A_{\text{rot}} x_{\text{rot}}, \qquad A_{\text{rot}} = \begin{bmatrix} 0_{3\times 3} & I_3 \\ 0_{3\times 3} & 0_{3\times 3} \end{bmatrix}. \tag{55}$$

The characteristic polynomial is again

$$\det(\lambda I - A_{\text{rot}}) = \lambda^6,$$

and therefore the eigenvalues are

$$\lambda_i = 0, \qquad i = 1, \ldots, 6.$$

Conclusion (Rotational Dynamics). The rotational subsystem also consists of three independent double integrators. Therefore:

- It is not asymptotically stable.
- It is not exponentially stable.
- Small angular velocity perturbations lead to linearly growing attitude errors.

4.3 Overall Hover Stability

Since both subsystems have eigenvalues at the origin, the full linearized hover model has twelve eigenvalues at zero. Thus, the open-loop hover equilibrium is **semistable** and not asymptotically stable.

In practice, this means:

- Any small disturbance in position or velocity leads to drift.
- Any small disturbance in attitude or angular rate leads to growing orientation error.
- Active feedback control is necessary to stabilize hover.

5 Connection to Project 01

Project 01 demonstrates that

- Open-loop trajectory tracking fails under perturbations.
- Small initial errors lead to trajectory deviation.
- Feedforward control is not robust.

This motivates closed-loop stabilization and feedback design.

6 Controllability of Linear Systems

Consider the linear time-invariant system

$$\dot{x} = Ax + Bu, \tag{56}$$

where $x \in \mathbb{R}^n$ and $u \in \mathbb{R}^m$.

6.1 Definition

The pair (A, B) is said to be **controllable** if, for any initial state $x(0) = x_0$ and any desired final state x_f, there exists an input $u(t)$ defined on $[0, T]$ that transfers the state from x_0 to x_f in finite time.

Equivalently, the system is controllable if the input can influence every state direction.

6.2 Controllability Matrix

Define the **controllability matrix**

$$\mathcal{C} \triangleq \begin{bmatrix} B & AB & A^2B & \cdots & A^{n-1}B \end{bmatrix}. \tag{57}$$

The system (56) is controllable if and only if

$$\operatorname{rank}(\mathcal{C}) = n. \tag{58}$$

6.3 Interpretation

- If (A, B) is controllable, then state feedback of the form

  $$u = Kx$$

  can arbitrarily assign the eigenvalues of $A + BK$.

- If the system is not controllable, then some internal modes cannot be influenced by the input and cannot be stabilized by feedback.

7 Controllability of Translational and Rotational Subsystems

We now check controllability of the linearized hover dynamics by examining the translational and rotational subsystems separately.

7.1 Translational Subsystem

Recall the translational state

$$x_{\text{tr}} \triangleq \begin{bmatrix} \delta p \\ \delta \dot{p} \end{bmatrix} \in \mathbb{R}^6,$$

and define the translational input

$$u_{\text{tr}} \triangleq \begin{bmatrix} \delta T \\ \delta q \end{bmatrix} = \begin{bmatrix} \delta T \\ \delta \psi \\ \delta \theta \\ \delta \phi \end{bmatrix} \in \mathbb{R}^4.$$

From (33), the matrices are

$$A_{\rm tr} = \begin{bmatrix} 0_{3\times3} & I_3 \\ 0_{3\times3} & 0_{3\times3} \end{bmatrix}, \qquad B_{\rm tr} = \begin{bmatrix} 0_{3\times1} & 0_{3\times3} \\ \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} & \begin{bmatrix} 0 & -g & 0 \\ 0 & 0 & g \\ 0 & 0 & 0 \end{bmatrix} \end{bmatrix}.$$

Rank test. The subsystem is controllable if and only if the controllability matrix

$$\mathcal{C}_{\rm tr} = \begin{bmatrix} B_{\rm tr} & A_{\rm tr}B_{\rm tr} & \cdots & A_{\rm tr}^5 B_{\rm tr} \end{bmatrix}$$

has rank 6.

Since $A_{\rm tr}$ is a collection of double integrators, it suffices to verify that each acceleration channel is influenced by at least one input:

$$\delta\ddot{p}_1 = -g\,\delta\theta, \tag{59}$$

$$\delta\ddot{p}_2 = g\,\delta\phi, \tag{60}$$

$$\delta\ddot{p}_3 = -\delta T. \tag{61}$$

Thus, the inputs $\delta\theta$, $\delta\phi$, and δT independently actuate the three translational accelerations, and hence each of the three double-integrator chains is controllable.

Conclusion (Translational Subsystem). The translational subsystem $(A_{\rm tr}, B_{\rm tr})$ is controllable, and in particular $\delta\psi$ does not affect translation at first order.

7.2 Rotational Subsystem

Recall the rotational state

$$x_{\rm rot} \triangleq \begin{bmatrix} \delta q \\ \delta\omega \end{bmatrix} \in \mathbb{R}^6, \qquad u_{\rm rot} \triangleq \delta\tau \in \mathbb{R}^3.$$

From (39), the matrices are

$$A_{\rm rot} = \begin{bmatrix} 0_{3\times3} & I_3 \\ 0_{3\times3} & 0_{3\times3} \end{bmatrix}, \qquad B_{\rm rot} = \begin{bmatrix} 0_{3\times3} \\ J^{-1} \end{bmatrix}.$$

Rank test. The controllability matrix

$$\mathcal{C}_{\rm rot} = \begin{bmatrix} B_{\rm rot} & A_{\rm rot}B_{\rm rot} & \cdots & A_{\rm rot}^5 B_{\rm rot} \end{bmatrix}$$

has rank 6 provided J is nonsingular (i.e., $J_1, J_2, J_3 > 0$). Equivalently, since

$$\delta\ddot{q} = J^{-1}\delta\tau,$$

each Euler-angle channel is a double integrator driven by its corresponding torque input.

Conclusion (Rotational Subsystem). If J is positive definite, then the rotational subsystem $(A_{\rm rot}, B_{\rm rot})$ is controllable.

7.3 Remark

Although the translational subsystem is controllable when treating $(\delta T, \delta q)$ as inputs, in the physical multicopter the attitude angles δq are not direct control inputs; they are generated by the rotational dynamics through $\delta\tau$. This motivates analyzing the controllability of the full 12-state model with inputs $(\delta T, \delta\tau)$ and the design of cascaded control laws.

8 LQR Design for Linear Stabilization

Consider the linear time-invariant system

$$\dot{x} = Ax + Bu, \tag{62}$$

where (A, B) is controllable.

8.1 Quadratic Performance Index

The Linear Quadratic Regulator (LQR) seeks a state-feedback control law

$$u = Kx \tag{63}$$

that minimizes the quadratic cost functional

$$J = \int_0^\infty \left(x^{\mathrm{T}}Qx + u^{\mathrm{T}}Ru\right)\, dt, \tag{64}$$

where

- $Q \in \mathbb{R}^{n\times n}$ is symmetric positive semidefinite ($Q \succeq 0$),
- $R \in \mathbb{R}^{m\times m}$ is symmetric positive definite ($R \succ 0$).

The matrix Q penalizes state deviations, while R penalizes control effort.

8.2 Algebraic Riccati Equation

The optimal gain matrix is given by

$$K = -R^{-1}B^{\mathrm{T}}P, \tag{65}$$

where P is the unique positive definite solution of the **continuous-time algebraic Riccati equation (CARE)**

$$A^{\mathrm{T}}P + PA - PBR^{-1}B^{\mathrm{T}}P + Q = 0. \tag{66}$$

8.3 Closed-Loop Stability

If (A, B) is controllable and $(A, Q^{1/2})$ is observable, then:

- The Riccati equation admits a unique positive definite solution P.
- The closed-loop system

$$\dot{x} = (A + BK)x \tag{67}$$

is asymptotically (indeed exponentially) stable.

Thus, although the open-loop hover dynamics are not stable, the LQR controller guarantees stabilization of the linearized system.

9 Pole Placement Using `place`

For a controllable linear time-invariant system

$$\dot{x} = Ax + Bu, \tag{68}$$

state feedback of the form

$$u = Kx \tag{69}$$

yields the closed-loop dynamics

$$\dot{x} = (A + BK)x. \tag{70}$$

If the pair (A, B) is controllable, then the eigenvalues of $(A + BK)$ can be assigned arbitrarily using pole placement.

9.1 Pole Placement Principle

Let $\{\lambda_1, \dots, \lambda_n\}$ be a desired set of closed-loop eigenvalues (with negative real parts for stability). There exists a gain matrix K such that

$$\mathrm{eig}(A + BK) = \{\lambda_1, \dots, \lambda_n\}. \tag{71}$$

The desired poles determine:

- Stability (real parts < 0),
- Convergence rate (distance from the imaginary axis),
- Oscillatory behavior (complex conjugate pairs),
- Damping ratio and natural frequency.

9.2 Using `place` in MATLAB

In MATLAB, the gain matrix can be computed using

```
K = -place(A,B,p)
```

where `p` is a vector of desired eigenvalues.

For example,

```
p = [-2 -2.5 -3 -3.5];
K = -place(A,B,p);
```

The resulting controller is

$$u = Kx, \tag{72}$$

and the closed-loop system matrix is $A_{\mathrm{cl}} = A + BK$.

9.3 Application to the Multicopter

Since the linearized hover model is controllable:

- The rotational poles can be placed to achieve fast attitude stabilization.
- The translational poles can be placed for slower, well-damped position regulation.
- The full 12-state model can be stabilized using a single gain matrix.

Pole placement provides direct eigenvalue assignment, whereas LQR provides an optimal trade-off between performance and control effort.

10 LQR Stabilization of Translational and Rotational Dynamics

Since both the translational and rotational subsystems are controllable but not open-loop stable, we can use LQR to stabilize each subsystem.

10.1 LQR for the Rotational Subsystem

Recall the rotational dynamics

$$\dot{x}_{\mathrm{rot}} = A_{\mathrm{rot}} x_{\mathrm{rot}} + B_{\mathrm{rot}} u_{\mathrm{rot}}, \tag{73}$$

where

$$x_{\mathrm{rot}} = \begin{bmatrix} \delta q \\ \delta\omega \end{bmatrix} \in \mathbb{R}^6, \qquad u_{\mathrm{rot}} = \delta\tau \in \mathbb{R}^3.$$

Choose weighting matrices

$$Q_{\mathrm{rot}} \succeq 0, \qquad R_{\mathrm{rot}} \succ 0,$$

and solve the algebraic Riccati equation

$$A_{\mathrm{rot}}^{\mathrm{T}} P_{\mathrm{rot}} + P_{\mathrm{rot}} A_{\mathrm{rot}} - P_{\mathrm{rot}} B_{\mathrm{rot}} R_{\mathrm{rot}}^{-1} B_{\mathrm{rot}}^{\mathrm{T}} P_{\mathrm{rot}} + Q_{\mathrm{rot}} = 0. \tag{74}$$

The optimal torque law is

$$\delta\tau = K_{\mathrm{rot}} x_{\mathrm{rot}}, \qquad K_{\mathrm{rot}} = -R_{\mathrm{rot}}^{-1} B_{\mathrm{rot}}^{\mathrm{T}} P_{\mathrm{rot}}. \tag{75}$$

Since $(A_{\mathrm{rot}}, B_{\mathrm{rot}})$ is controllable, the closed-loop system

$$\dot{x}_{\mathrm{rot}} = (A_{\mathrm{rot}} + B_{\mathrm{rot}} K_{\mathrm{rot}}) x_{\mathrm{rot}}$$

is exponentially stable.

Interpretation. The rotational LQR stabilizes attitude and angular rates. This forms the inner-loop controller.

10.2 LQR for the Translational Subsystem

The translational dynamics are

$$\dot{x}_{\mathrm{tr}} = A_{\mathrm{tr}} x_{\mathrm{tr}} + B_{\mathrm{tr}} u_{\mathrm{tr}}, \tag{76}$$

where

$$x_{\mathrm{tr}} = \begin{bmatrix} \delta p \\ \delta \dot{p} \end{bmatrix} \in \mathbb{R}^6, \qquad u_{\mathrm{tr}} = \begin{bmatrix} \delta T \\ \delta q \end{bmatrix}.$$

Choose weighting matrices

$$Q_{\mathrm{tr}} \succeq 0, \qquad R_{\mathrm{tr}} \succ 0,$$

and solve the Riccati equation

$$A_{\mathrm{tr}}^{\mathrm{T}} P_{\mathrm{tr}} + P_{\mathrm{tr}} A_{\mathrm{tr}} - P_{\mathrm{tr}} B_{\mathrm{tr}} R_{\mathrm{tr}}^{-1} B_{\mathrm{tr}}^{\mathrm{T}} P_{\mathrm{tr}} + Q_{\mathrm{tr}} = 0. \tag{77}$$

The optimal feedback is

$$u_{\mathrm{tr}} = K_{\mathrm{tr}} x_{\mathrm{tr}}, \qquad K_{\mathrm{tr}} = -R_{\mathrm{tr}}^{-1} B_{\mathrm{tr}}^{\mathrm{T}} P_{\mathrm{tr}}. \tag{78}$$

Interpretation. The translational LQR computes desired thrust and attitude corrections to regulate position and velocity.

10.3 Full-System or Cascaded Design

There are two standard approaches:

- **Full-state LQR:** Design a single gain

  $$u = Kx,$$

  for the complete 12-state model.

- **Cascaded LQR:**
  - Inner loop: stabilize (q, ω) using τ.
  - Outer loop: regulate $(p, \dot{p})$ by generating desired (T, q).

Both approaches stabilize hover. The cascaded approach reflects the physical structure of the multicopter and is widely used in practice.

11 Why Is Integral Action Needed?

State feedback can stabilize the linearized hover model, but stabilization alone does not guarantee accurate regulation in the presence of disturbances or modeling error. To understand why integral action is necessary, consider the following.

11.1 Constant Disturbance Example

Suppose the linearized system is subject to a constant disturbance:

$$\dot{x} = Ax + Bu + Ew, \tag{79}$$

where w is constant (e.g., steady wind).

Using pure state feedback,

$$u = -Kx, \tag{80}$$

the closed-loop dynamics become

$$\dot{x} = (A - BK)x + Ew. \tag{81}$$

If $(A - BK)$ is Hurwitz, the steady-state satisfies

$$0 = (A - BK)x_{\text{ss}} + Ew, \tag{82}$$

which implies

$$x_{\text{ss}} = -(A - BK)^{-1}Ew. \tag{83}$$

In general, $x_{\text{ss}} \neq 0$.

Conclusion: Pure state feedback stabilizes the system but does not eliminate steady-state error caused by constant disturbances.

11.2 Why Proportional Feedback Is Insufficient

State feedback is proportional to the state:

$$u \propto x.$$

If a disturbance produces a constant position offset, the controller applies a constant correcting input proportional to that offset. Equilibrium is reached when control balances disturbance, but the position error remains nonzero.

Thus, proportional feedback alone cannot drive steady-state error to zero.

11.3 Role of the Integrator

Introduce the integral state

$$\eta(t) = \int_0^t e_p(\tau)\, d\tau, \tag{84}$$

with dynamics

$$\dot{\eta} = e_p.$$

If the position error remains nonzero, the integral state continues to grow. The control law

$$u = -K_x x - K_i \eta \tag{85}$$

therefore keeps increasing the corrective input until the error is forced to zero.

The only steady-state compatible with bounded signals is

$$e_p = 0.$$

11.4 Fundamental Principle

To reject a constant disturbance, the controller must contain an internal model of a constant signal. An integrator provides exactly this model.

Key Insight. Stability ensures bounded motion. Integral action is required to guarantee zero steady-state error in the presence of constant disturbances. Without it, hover may be stable but inaccurate.

12 Full-State Feedback with Integral Action

In addition to pure state feedback (e.g., LQR), improved trajectory tracking can be achieved by augmenting the linearized hover model with integral action. This ensures zero steady-state error in the presence of constant disturbances or modeling uncertainty.

12.1 Augmented State

Consider the linearized multicopter model

$$\dot{x} = Ax + Bu, \tag{86}$$

where $x \in \mathbb{R}^{12}$ and $u \in \mathbb{R}^4$.

Let the tracking error for the position states be

$$e_p = p - p_d, \tag{87}$$

where p_d is the desired position trajectory.

Define integral states

$$\eta(t) \triangleq \int_0^t e_p(\tau)\, d\tau, \tag{88}$$

with dynamics

$$\dot{\eta} = e_p. \tag{89}$$

The augmented state becomes

$$x_a \triangleq \begin{bmatrix} x \\ \eta \end{bmatrix} \in \mathbb{R}^{15}. \tag{90}$$

12.2 Augmented Dynamics

The augmented system can be written as

$$\dot{x}_a = A_a x_a + B_a u + E_a p_d, \tag{91}$$

where

$$A_a = \begin{bmatrix} A & 0 \\ C_p & 0 \end{bmatrix}, \qquad B_a = \begin{bmatrix} B \\ 0 \end{bmatrix}, \tag{92}$$

and C_p extracts the position states from x, that is,

$$C_p x = p.$$

12.3 Integral State Feedback Law

A full-state feedback with integral action has the form

$$u = K_x x + K_i \eta, \tag{93}$$

or equivalently,

$$u = K_a x_a, \qquad K_a = \begin{bmatrix} K_x & K_i \end{bmatrix}. \tag{94}$$

The gains (K_x, K_i) can be obtained using pole placement or LQR applied to the augmented system (A_a, B_a).

12.4 Interpretation

- The state feedback term $K_x x$ stabilizes the fast translational and rotational dynamics.
- The integral term $K_i \eta$ eliminates steady-state position error.
- For constant disturbances, the integral action ensures $e_p(t) \to 0$.

Thus, full-state feedback with integral augmentation provides stabilization and accurate trajectory tracking around hover.

13 Time-Scale Separation Principle and Cascaded Control

The linearized multicopter dynamics at hover exhibit a natural separation between

- **Rotational (attitude) dynamics**, and
- **Translational (position) dynamics**.

13.1 Physical Observation

From the linearized model,

$$\delta \ddot{p}_1 = -g\,\delta\theta, \tag{95}$$
$$\delta \ddot{p}_2 = g\,\delta\phi, \tag{96}$$
$$\delta \ddot{p}_3 = -\delta T, \tag{97}$$

while

$$\delta \ddot{q} = J^{-1} \delta\tau. \tag{98}$$

The rotational dynamics are directly actuated by torques and depend only on inertia. The translational dynamics are indirectly actuated through attitude angles and thrust.

In practice:

- Attitude dynamics are fast (small inertia, high torque authority).
- Position dynamics are slow (double integration of acceleration).

13.2 Time-Scale Separation Assumption

Let the rotational closed-loop dynamics be

$$\dot{x}_{\text{rot}} = (A_{\text{rot}} - B_{\text{rot}} K_{\text{rot}}) x_{\text{rot}}.$$

If the rotational controller is designed such that its eigenvalues satisfy

$$|\lambda_{\text{rot}}| \gg |\lambda_{\text{tr}}|,$$

then the rotational subsystem converges much faster than the translational subsystem.

Under this assumption,

- The attitude dynamics can be approximated as instantaneous.
- The translational subsystem can treat (ϕ, θ) as virtual control inputs.

This is the **time-scale separation principle**.

13.3 Justification of Cascaded Control

Time-scale separation justifies the cascaded architecture:

- **Inner loop (fast):** Stabilize (q, ω) using τ.
- **Outer loop (slow):** Regulate $(p, \dot{p})$ by commanding desired (T, q).

Mathematically, if the inner loop is sufficiently fast and stable, then the outer-loop design can be performed using the reduced-order model

$$\delta\ddot{p} = \begin{bmatrix} -g\,\theta \\ g\,\phi \\ -\delta T \end{bmatrix},$$

where (ϕ, θ) are treated as control inputs.

13.4 Practical Implication

In multicopter design:

- Attitude bandwidth is typically 5–10 times larger than position bandwidth.
- This ensures negligible interaction between the loops.
- The overall closed-loop system behaves as if the inner loop were ideal.

Thus, time-scale separation provides both physical intuition and mathematical justification for the widely used inner–outer loop cascaded controller architecture.

14 Validity of Linearization

The linearized hover model was obtained using a first-order Taylor expansion of the nonlinear dynamics about the equilibrium (x_e, u_e). It is therefore a *local approximation* that is valid only in a neighborhood of hover.

14.1 Small-Angle Assumption

The key approximation used in the translational linearization was

$$\mathcal{O}_{A/D}(q)e_3 \approx \begin{bmatrix} \theta \\ -\phi \\ 1 \end{bmatrix}, \tag{99}$$

which relies on

$$\sin(\phi) \approx \phi, \qquad \sin(\theta) \approx \theta, \qquad \cos(\phi) \approx 1, \qquad \cos(\theta) \approx 1.$$

These approximations are accurate only when

$$|\phi|,\ |\theta| \ll 1 \text{ rad}.$$

In practice, the linear hover model is typically accurate for attitude angles up to approximately 10°–15°.

14.2 Local Nature of the Linear Model

Let

$$\dot{x} = f(x, u)$$

be the nonlinear multicopter dynamics and

$$\dot{\xi} = A\xi + Bv$$

be the linearized perturbation model. The linear model neglects higher-order terms of the form

$$\mathcal{O}(\|\xi\|^2,\ \|\xi\|\,\|v\|).$$

Thus:

- For small perturbations, the linear model accurately predicts behavior.
- For large attitude angles or aggressive maneuvers, higher-order nonlinear terms become significant.
- Far from hover, the linearized model may give incorrect predictions of stability and performance.

14.3 Implications for Control Design

- LQR or pole-placement gains designed using the hover model guarantee stability only locally.
- Large commanded accelerations require large roll/pitch angles, which may invalidate the small-angle approximation.
- Aggressive flight (e.g., flips, high-speed forward flight) requires nonlinear control methods.

14.4 Engineering Interpretation

The linearized hover model is appropriate for:

- Stabilization about hover,
- Slow trajectory tracking,
- Small position corrections,
- Controller bandwidth design.

It is not appropriate for:

- Large-angle maneuvers,
- Inverted flight,
- Aerodynamic effects at high forward speed,
- Strongly nonlinear regimes.

Key Insight. Linearization provides a powerful and accurate description of multicopter dynamics *near hover*. However, its validity is local, and control performance outside this region must be verified using the full nonlinear model.

15 Actuator Saturation and Integrator Windup

The linear control designs developed in previous sections (pole placement, LQR, integral augmentation) assume that the control input u can take arbitrary values. In a physical multicopter, this assumption is not valid.

15.1 Actuator Saturation

The inputs

$$u = \begin{bmatrix} T \\ \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix}$$

are physically limited:

- Thrust satisfies $0 \le T \le T_{\max}$.
- Torques satisfy $|\tau_i| \le \tau_{i,\max}$.

In practice, the commanded control signal

$$u_c = K_x x + K_i \eta$$

is passed through a saturation nonlinearity:

$$u = \mathrm{sat}(u_c).$$

Thus, even if the controller computes a large corrective input, the actuator may be unable to realize it.

15.2 What Is Integrator Windup?

Recall that the integral state satisfies

$$\dot{\eta} = e_p.$$

Suppose a large disturbance causes a significant position error. The controller responds by increasing u_c through the integral term. If u_c exceeds actuator limits, the actual input u saturates:

$$u \neq u_c.$$

However, the integrator continues to integrate the error:

$$\dot{\eta} = e_p \neq 0.$$

As a result:

- The integral state grows large,
- The commanded input u_c becomes very large,
- Once the system re-enters the controllable region, excessive control effort may occur,
- Overshoot and oscillations can result.

This phenomenon is called **integrator windup**.

15.3 Why Windup Occurs

Windup occurs because the integrator assumes that the commanded input is applied exactly. When saturation occurs, the internal controller model does not match the physical actuator.

In other words:

$$\text{Controller assumption: } u = u_c,$$

$$\text{Reality: } u = \text{sat}(u_c).$$

This mismatch breaks the linear closed-loop analysis.

15.4 Anti-Windup Concept

To mitigate windup, practical controllers modify the integrator when saturation occurs.

Common strategies include:

- **Integrator clamping:** Stop integrating when the actuator is saturated and the error would increase saturation.
- **Back-calculation:** Add a correction term proportional to $(u - u_c)$:

  $$\dot{\eta} = e_p + K_{aw}(u - u_c),$$

  where K_{aw} is an anti-windup gain.

These modifications prevent the unbounded growth of the integral state during actuator saturation.

15.5 Engineering Implication

Integral action is necessary for disturbance rejection, but actuator limits are unavoidable in multicopters. Therefore:

- Integral gains must be chosen carefully,
- Saturation must be considered in simulation,
- Anti-windup mechanisms are required in flight controllers.

Key Insight. Linear control design assumes unlimited actuation. Real multicopters do not satisfy this assumption. When integral action is combined with saturation, nonlinear effects such as windup must be addressed explicitly.

16 Conceptual Questions for Discussion

1. **Open-Loop Hover Behavior.**

   The linearized multicopter model at hover has twelve eigenvalues at zero. Why does the vehicle not "blow up" immediately in open loop? Is the hover equilibrium unstable, asymptotically stable, or Lyapunov stable?

   Answer. Eigenvalues at zero correspond to integrators, not exponentially unstable modes. If the zero eigenvalues are simple (no Jordan blocks), the system is *Lyapunov stable* but not asymptotically stable. Physically, small perturbations do not grow exponentially; instead, velocity perturbations lead to position drift and angular-rate perturbations lead to linearly growing attitude errors. The system drifts but does not "explode."

2. **Yaw and Translation Coupling.**

   Why does the yaw angle ψ not appear in the linearized translational dynamics at hover?

   Answer. At hover and under the small-angle approximation, the thrust direction depends on roll ϕ and pitch θ to first order, but not on yaw ψ. Yaw corresponds to rotation about the vertical axis, which does not change the vertical thrust direction to first order. Therefore, ψ does not affect translational accelerations in the linearized model.

3. **Time-Scale Separation.**

   Suppose the inner-loop attitude controller is designed with very slow closed-loop poles. What effect would this have on the outer-loop position controller?

   Answer. The outer-loop controller assumes that the inner-loop attitude dynamics are fast and track commanded angles almost instantaneously. If the inner loop is slow, this assumption breaks down. The translational controller would command attitudes that are not achieved quickly, introducing additional phase lag and coupling. This can lead to oscillations or instability. Time-scale separation ensures that the inner loop behaves approximately as an ideal actuator for the outer loop.

4. **Loss of Actuation and Controllability.**

   Assume that the pitch torque input τ_2 fails (i.e., $\tau_2 = 0$ for all time). Is the linearized multicopter system still controllable near hover?

   Answer. No. Without τ_2, the pitch angle θ cannot be directly controlled. Since forward acceleration $\ddot{p}_1$ depends on θ, the vehicle loses the ability to regulate motion in that direction. Thus, the system becomes partially uncontrollable and cannot independently regulate all translational directions.

5. **Pole Placement vs. LQR.**

   Both pole placement and LQR can stabilize the linearized hover model. Why might LQR be preferred in practice?

   Answer. LQR provides a systematic method for balancing performance and control effort through the weighting matrices Q and R. It avoids arbitrary pole selection, often results in better-conditioned gains, and naturally handles multi-input systems. Pole placement directly assigns eigenvalues but does not explicitly penalize control effort.

6. **Underactuation.**

   The multicopter has six degrees of freedom but only four control inputs. Why is it considered an underactuated system, even though the linearized hover model is controllable?

   Answer. The vehicle has six configuration variables (three position and three orientation) but only four independent inputs (thrust and three torques). Translational motion is indirectly controlled through attitude changes. Controllability means the state can be driven to any nearby state over time, but underactuation means not all accelerations can be independently commanded at an instant. Thus, the system is controllable but not fully actuated.